

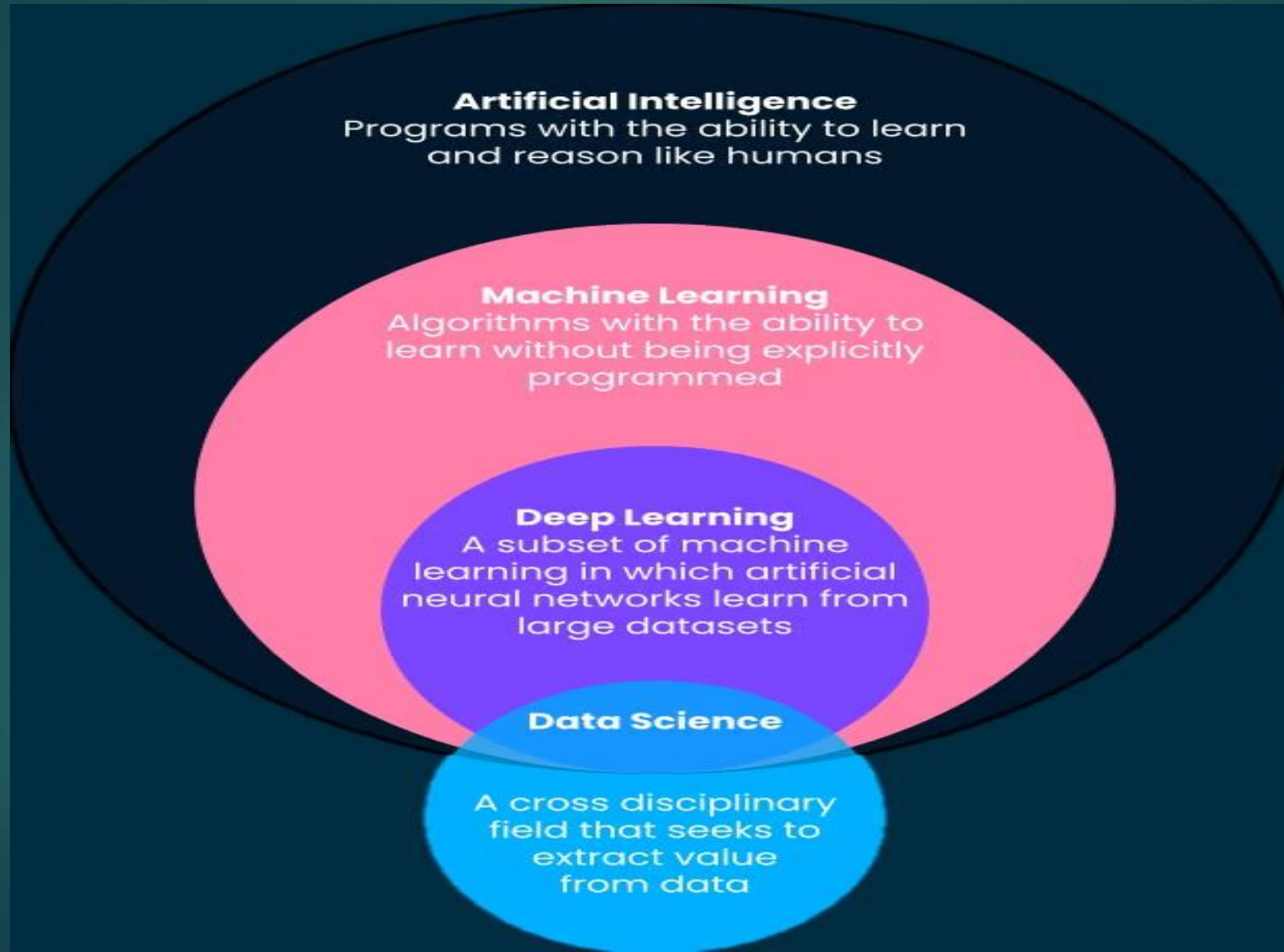


Machine Learning

BY

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Machine learning vs. AI vs. deep learning



- **AI** refers to the development of programs that behave intelligently and mimic human intelligence through a set of algorithms. The field focuses on three skills: learning, reasoning, and self-correction to obtain maximum efficiency. AI can refer to either machine learning-based programs or even explicitly programmed computer programs.
- **Machine learning** is a subset of AI, which uses algorithms that learn from data to make predictions. These predictions can be generated through supervised learning, where algorithms learn patterns from existing data, or unsupervised learning, where they discover general patterns in data. ML models can predict numerical values based on historical data, categorize events as true or false, and cluster data points based on commonalities.
- **Deep learning**, on the other hand, is a subfield of machine learning dealing with algorithms based essentially on multi-layered artificial neural networks (ANN) that are inspired by the structure of the human brain.

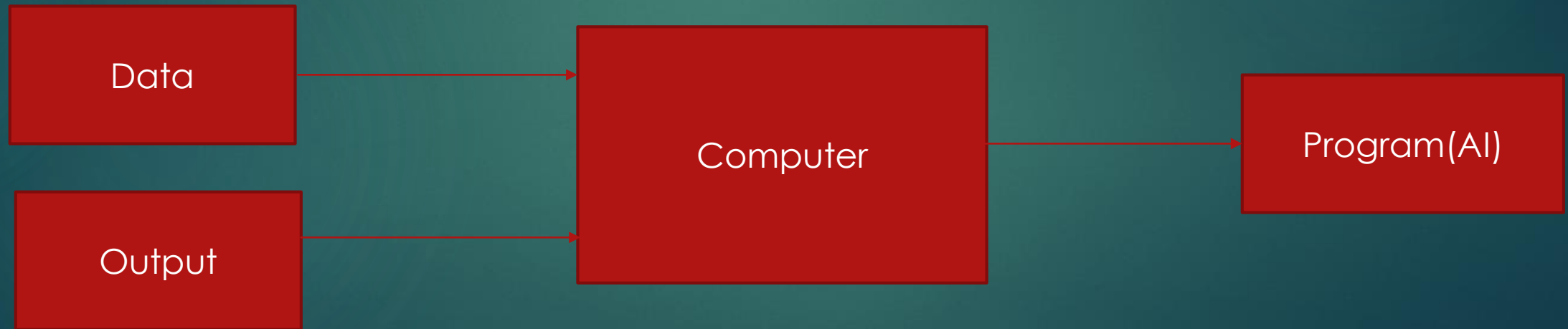
From Traditional Programming to Machine Learning

In traditional programming, a computer follows a set of predefined instructions to perform a task. However, in machine learning, the computer is given a set of examples (data) and a task to perform, but it's up to the computer to figure out how to accomplish the task based on the examples it's given.



Traditional Programming to Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that enables computers to learn from data and make decisions or predictions without being explicitly programmed. In ML, systems improve their performance over time as they are exposed to more data.



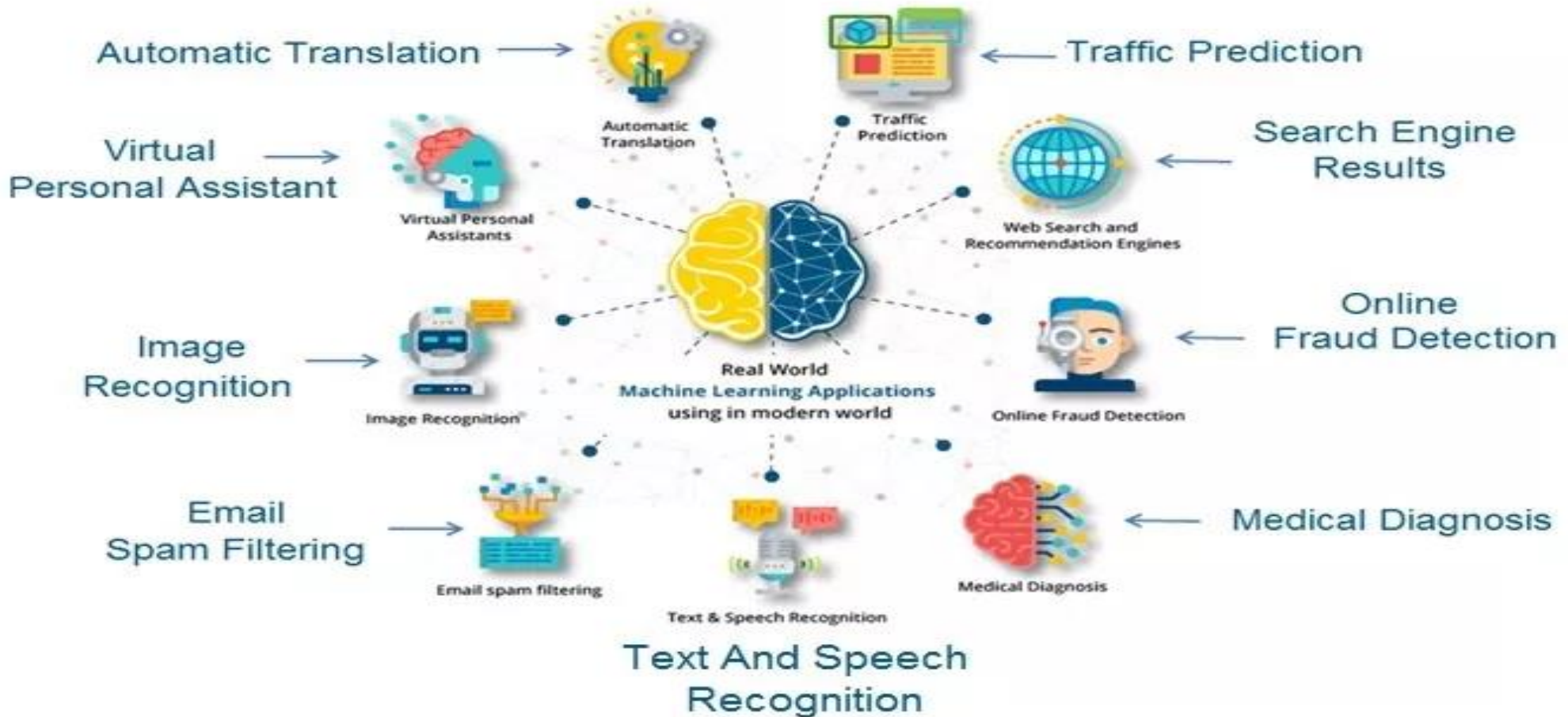


For instance, if we want a computer to recognize images of cats, we don't provide it with specific instructions on what a cat looks like. Instead, we give it thousands of images of cats and let the machine learning algorithm figure out the common patterns and features that define a cat. Over time, as the algorithm processes more images, it gets better at recognizing cats, even when presented with images it has never seen before.

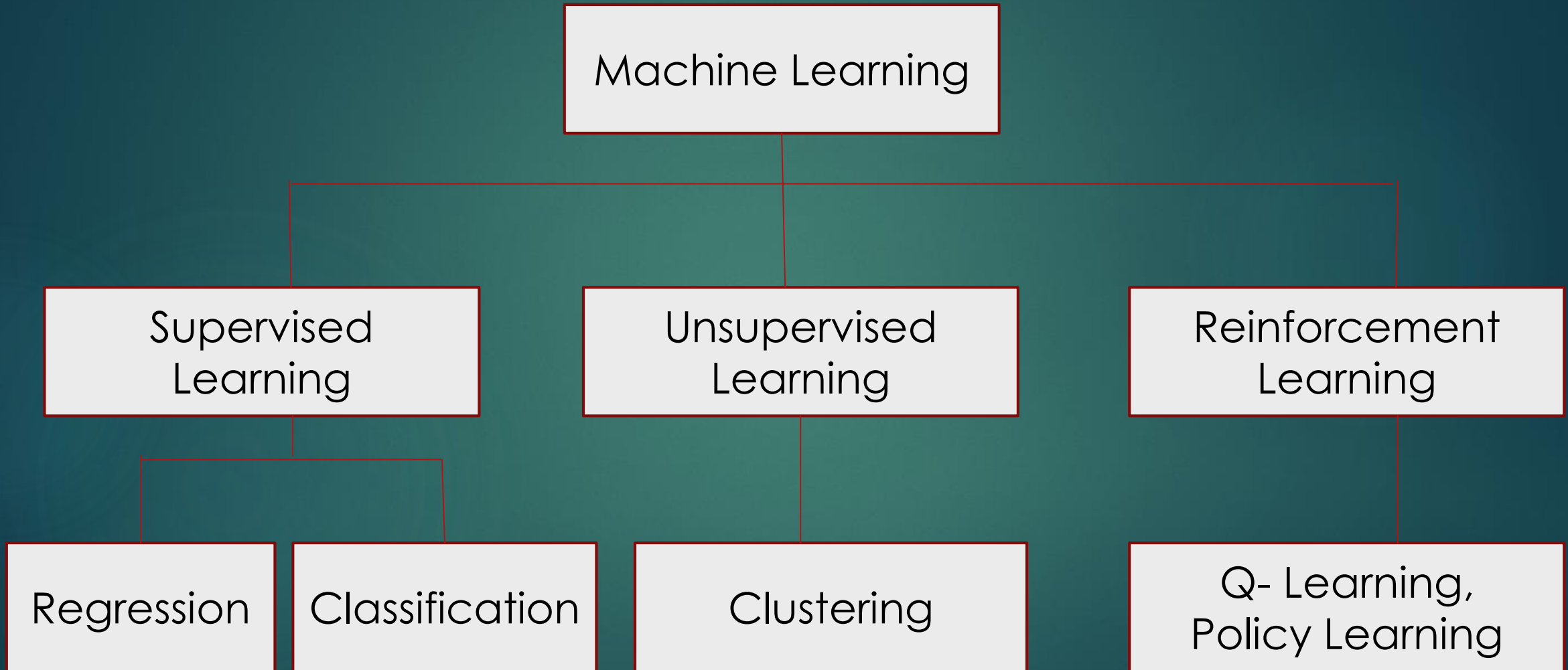
This ability to learn from data and improve over time makes machine learning incredibly powerful and versatile. It's the driving force behind many of the technological advancements we see today, from voice assistants and recommendation systems to self-driving cars and predictive analytics.

Usage of Machine Learning

Top Real-World Examples of Machine Learning



Types Of Machine Learning

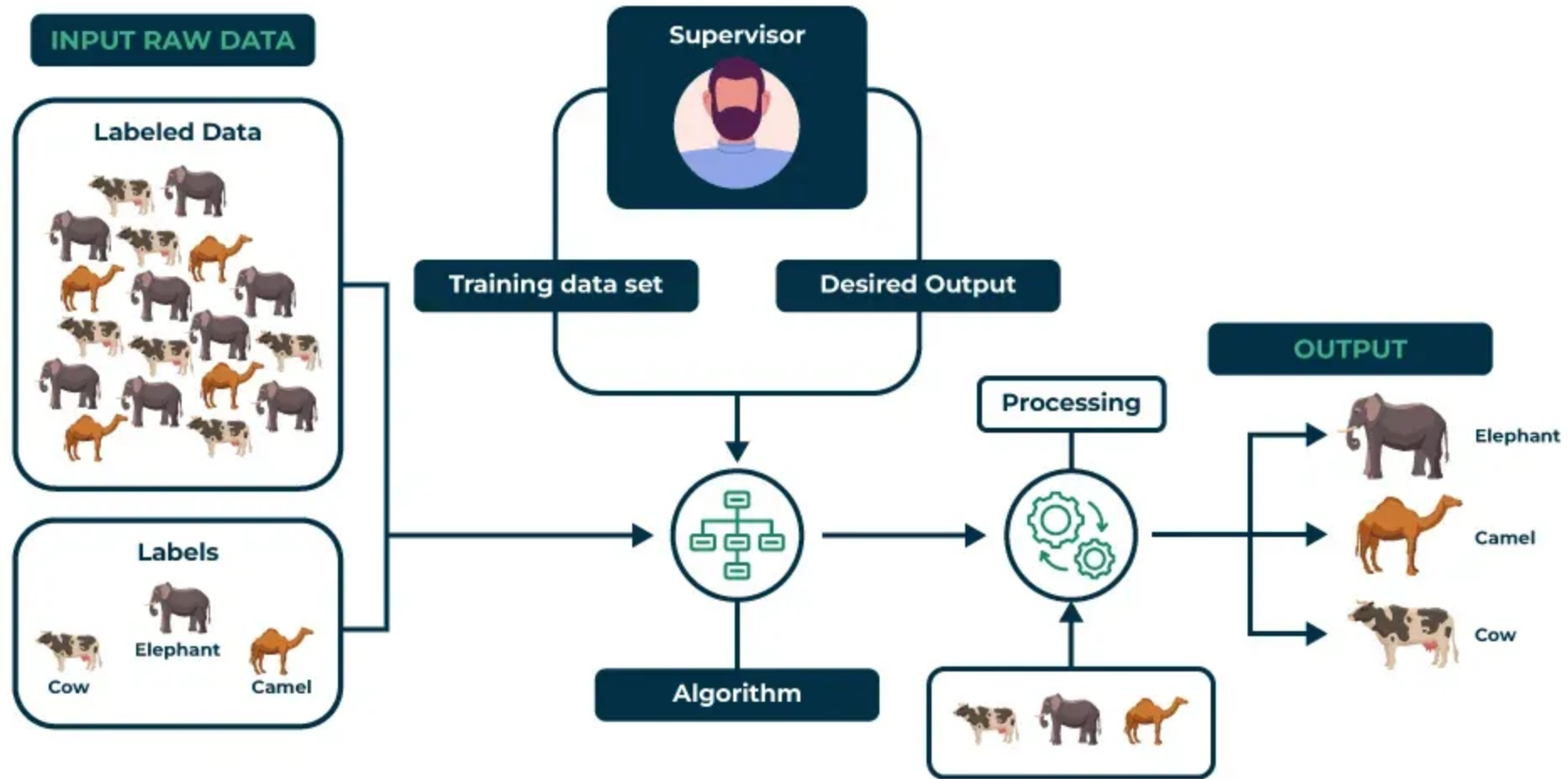


Supervised Learning

In Supervised learning, an AI system is presented with data which is labeled, which means that each data is tagged with the correct label.

The goal is to approximate the mapping function so well that when you have new input data (x) that you can predict the output variables (Y) for that data.

Supervised Learning



Applications of Supervised Learning

Regression

- Real Estate Prediction
- Weather Forecasting
- Financial Portfolio Prediction

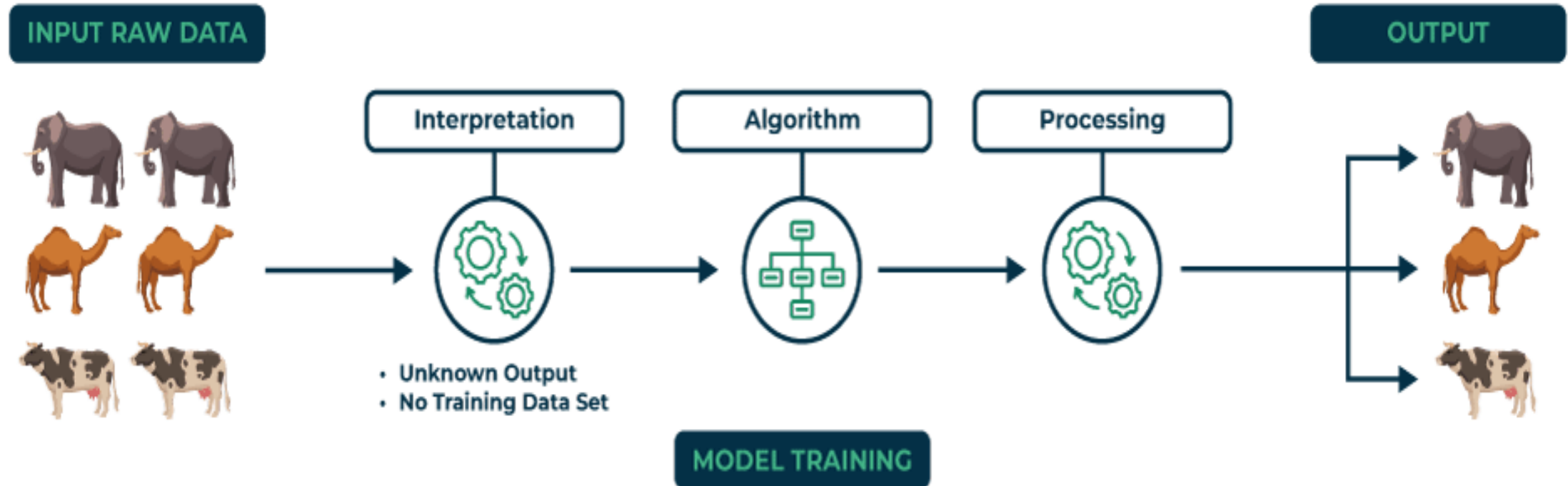
Classification

- Credit Card Fraud Detection
- Image Classification
- Spam Detection
- Insurance Decisioning

Unsupervised Learning

Unlike supervised learning, where the data is labeled with a specific category or outcome, unsupervised learning algorithms **are tasked with finding patterns and relationships within the data without any prior knowledge of the data's meaning**. Unsupervised machine learning algorithms **find hidden patterns and data without any human intervention, i.e., we don't give output to our model. The training model has only input parameter values and discovers the groups or patterns on its own.**

Unsupervised Learning



Types Of Supervised Learning

- **Classification:** Classification is a type of supervised machine learning that involves training a model to identify which category or class an input data point belongs to. In other words, classification is a way to automatically assign a label or category to a given piece of input data. A classification problem is when the output variable is a category, such as “red” or “blue” or “disease” and “no disease”.
- **Regression:** Regression is primarily used when we are trying to model continuous numerical response variable. A regression problem is when the output variable is a real value, such as “dollars” or “weight”.

Applications of Unsupervised Learning

Clustering

- Document theme extraction
- Customer Segmentation
- Insurance Fraud Detection
- Delivery Store Optimization

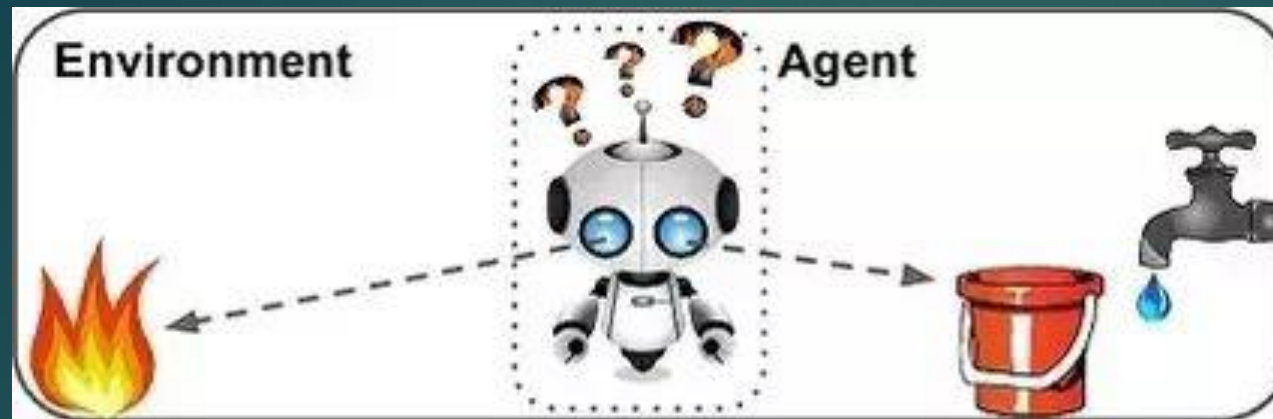
Reinforcement Learning

A reinforcement learning algorithm, or agent, learns by interacting with its environment. The agent receives rewards by performing correctly and penalties for performing incorrectly.

Unlike supervised and unsupervised learnings, reinforcement learning has a feedback type of algorithm. In other words, for every result obtained the algorithm gives **feedback** to the model under training

Reinforcement Learning in ML





1 Observe

2 Select action using policy



3 Action!

4 Get reward or penalty



5 Update policy (learning step)

6 Iterate until an optimal policy is found

Applications of Reinforcement Learning

- Traffic Light Control
- Resource Management
- Robotics
- Games
- Bidding and Advertisement

Steps in Supervised Machine Learning Model

